Day 21: Model comparison for logistic regression

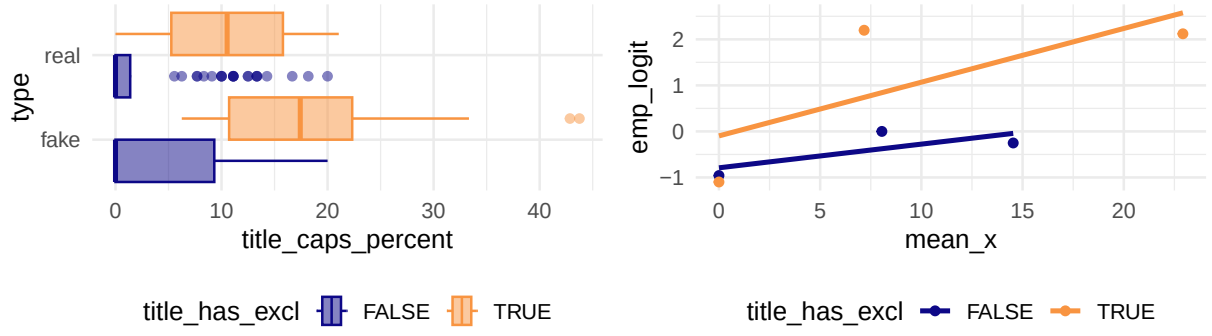
Last time we learned how to do inference for **one** coefficient (Wald test and CI) and how to check conditions. Today we compare **models**:

1. Do we need an interaction term?
2. The **likelihood ratio test** (LRT): the logistic version of the extra sums of squares F-test
3. AIC and BIC for logistic regression

The model from last time

term	estimate	std. error	z	p-value
Intercept	-2.232	0.672	-3.32	<0.001
title_words	0.120	0.058	2.07	0.038
title_has_exclTRUE	1.998	0.849	2.35	0.019
title_caps_percent	0.050	0.029	1.69	0.091

Question: is the association between `title_caps_percent` and the odds of being fake news *different* for titles with and without an exclamation point? That is, do we need an **interaction** between `title_has_excl` and `title_caps_percent`?



The model with an interaction

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 \text{ words}_i + \beta_2 \text{ excl}_i + \beta_3 \text{ caps}_i + \beta_4 (\text{excl}_i \times \text{caps}_i)$$

term	estimate	std. error	z	p-value
Intercept	-2.273	0.677	-3.36	<0.001
title_words	0.127	0.059	2.14	0.032

term	estimate	std. error	z	p-value
title_has_exclTRUE	1.281	1.386	0.92	0.355
title_caps_percent	0.041	0.033	1.28	0.202
title_has_exclTRUE:title_caps_percent	0.060	0.100	0.60	0.549

Q1. Write the fitted model on the log-odds scale for each group separately.

With an exclamation point ($excl = 1$):

Without an exclamation point ($excl = 0$):

Q2. Do we need an interaction term? Write out the null and alternative hypothesis.

Q3. Do we need `title_has_excl` at all? Write out the null and alternative hypothesis for this question.

The likelihood ratio test (LRT)

In linear regression we compared nested models with the _____ F-test. For logistic regression we use a **likelihood ratio test**.

Deviance

The **deviance** of a model is $-2 \log L$, where L is the maximized likelihood. Like SSE in OLS, a smaller deviance means a _____ fit. A bigger model always has a deviance that is the same or _____.

Likelihood Ratio Test

Hypotheses: H_0: _____ vs. H_a: at least one of these β's $\neq 0$
Test statistic: Λ = Dev(reduced) − Dev(full)
Reference distribution: χ^2 with df = _____
p-value: the area _____ Λ (upper tail)

In R, give the **reduced** model first:

```
anova(reduced_model, full_model, test = "LRT")
```

Resid DF	Resid Dev	Df	Deviance	Pr(>Chi)
$n-$ (p+1 in reduced)	Dev(reduced)			
$n-$ (p+1 in full)	Dev(full)	# extra coefs	Λ	p-value

Example 1: do we need the interaction?

```
Analysis of Deviance Table

Model 1: type_num ~ title_words + title_has_excl + title_caps_percent
Model 2: type_num ~ title_words + title_has_excl * title_caps_percent
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       146     170.86
2       145     170.44  1  0.41625   0.5188
```

Hypotheses:

Test statistic, df, and p-value:

Conclusion in context:

Example 2: the overall test

The summary() output gives the deviance of the model, and of an intercept-only model:

```
summary(news_mod)
```

```
Call:
glm(formula = type_num ~ title_words + title_has_excl + title_caps_percent,
    family = "binomial", data = news)

Coefficients:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)        -2.23184    0.67187  -3.322 0.000894 ***
title_words         0.12034    0.05808   2.072 0.038260 *
title_has_exclTRUE  1.99783    0.84918   2.353 0.018640 *
title_caps_percent  0.04982    0.02946   1.691 0.090783 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 201.90  on 149  degrees of freedom
Residual deviance: 170.86  on 146  degrees of freedom
AIC: 178.86

Number of Fisher Scoring iterations: 4
```

Q4. Fill in the ANOVA table below for the test $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ against H_A at least $1 \neq 0$

	Resid DF	Resid Dev	Df	Deviance	Pr(>Chi)
1					
2					

```
1 - pchisq(lambda, df)
```

Conclusion:

z-test vs. LRT

For one coefficient, $H_0 : \beta_i = 0$:

- In OLS, the t-test and F-test give *exactly* the same p-value.
- In logistic regression the z-test (Wald) and the LRT are _____ equivalent. Hopefully they are close. If not, the _____ is usually more accurate.

term	z-test p-value	LRT p-value
title_words	0.0383	0.0355
title_has_excl	0.0186	0.0075
title_caps_percent	0.0908	0.0869

The LRT can also test **several coefficients at once**, which the z-test cannot.

Example 3: is `title_has_excl` needed at all?

`title_has_excl` appears in two terms of `news_mod2`. To drop it entirely we have to remove both the main effect and the interaction.

H_0: _____

```
Analysis of Deviance Table

Model 1: type_num ~ title_words + title_caps_percent
Model 2: type_num ~ title_words + title_has_excl * title_caps_percent
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       147     178.01
2       145     170.44  2   7.5695  0.02271 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Q5. The individual z-tests for these two terms have large p-values. The joint LRT has a small p-value. Why? (Think back to the F-test in MLR.)

Conclusion in context:

AIC and BIC for logistic regression

$$AIC = -2\log(L) + 2k \qquad BIC = -2\log(L) + \log(n) \cdot k$$

where L is the maximized likelihood and k is the number of coefficients (including the intercept). For binary logistic regression, $-2\log(L)$ is the _____. Smaller is better, and both can compare nested **or** non-nested models.

	df	AIC	BIC
reduced_model	3	184.01	193.04
news_mod	4	178.86	190.90
news_mod2	5	180.44	195.50

Q6. Which model do AIC and BIC prefer? Do they agree with the LRTs?

Summary

To compare…	Use
one coefficient to 0	
several coefficients to 0, nested models	
non-nested models, or many models	

- Deviance is $-2\log L$. It plays the role of _____ in OLS.
- We use deviance to compare models

R function reference

Function	Purpose
`anova(reduced, full, test = "LRT")`	Likelihood ratio test for nested models
`deviance(model)`	Deviance ($-2\log L$) of a model
`pchisq(x, df)`	χ^2 probability; use `1 - pchisq()` for a p-value
`AIC(m1, m2, ...)`, `BIC(m1, m2, ...)`	Information criteria for one or more models